

Linear Assignment #103
 Name: Yash Raj
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Question #1.

(i)
$$\begin{bmatrix} \lambda-4 & -2 & -3 & -3 \\ 0 & \lambda-2 & -4 & -3 \\ 0 & 0 & \lambda-4 & -14 \\ 0 & 0 & 0 & \lambda-2 \end{bmatrix}$$
 (all of $\lambda \neq 4$)

Solⁿ

$$\lambda-4 \neq 0 \Rightarrow \begin{bmatrix} 0 & -2 & -3 & -3 \\ 0 & 2 & -\lambda & -3 \\ 0 & 0 & 0 & -14 \\ 0 & 0 & 0 & 2 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 0 & 2 & -\lambda & -3 \\ 0 & -2 & -3 & -3 \\ 0 & 0 & 0 & -14 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

$$\begin{matrix} R_1 + R_2 \\ R_3 + R_4 \\ R_3/2 \end{matrix} \begin{bmatrix} 0 & 2 & -\lambda & -3 \\ 0 & 0 & -3-\lambda & -6 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_4 = 0, x_1 = 3$$

$$2x_2 - 4x_3 - 3x_4 = 0$$

$$(-3-\lambda)x_3 - 6x_4 = 0$$

for all of $\lambda \neq 4$ to be 2 there should be 2 free-variables / non-pivot rows

c) R_1 & R_3 are pivots, R_4 is not so far

$R_3: -\lambda-3=0$ then λ will also

be non-pivot. $\lambda = -3 \Rightarrow -\lambda-3=0$ h=3

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Question #2

$$x_1 = -(x_2 + x_3 + x_4)$$

$$x_2 = s, x_3 = t, x_4 = u$$

$$(x_1, x_2, x_3, x_4) = s(-1, 1, 0, 0) + t(-1, 0, 1, 0)$$

$$\text{Basis} = \left\{ \underbrace{(-1, 1, 0, 0)}_{u_1}, \underbrace{(-1, 0, 1, 0)}_{u_2}, \underbrace{(-1, 0, 0, 1)}_{u_3} \right\}$$

$$v_1 = u_1 = (-1, 1, 0, 0)$$

$$v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1 = (-1, 0, 1, 0) - \frac{1}{2}(-1, 1, 0, 0)$$

$$v_2 = \left(-\frac{1}{2}, -\frac{1}{2}, 1, 0\right)$$

$$v_3 = u_3 - \frac{\langle u_3, v_2 \rangle}{\|v_2\|^2} v_2 - \frac{\langle u_3, v_1 \rangle}{\|v_1\|^2} v_1$$

$$= (-1, 0, 0, 1) - \frac{1}{3} \left(-\frac{1}{2}, -\frac{1}{2}, 1, 0\right) - \frac{1}{2}(-1, 1, 0, 0)$$

$$v_3 = \left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, 1\right) \times 3 = (-1, -1, -1, 3)$$

$$\text{Basis} = \left\{ (-1, 1, 0, 0), \left(-\frac{1}{2}, -\frac{1}{2}, 1, 0\right), \left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, 1\right) \right\}$$

$$\text{Simplified} = \left\{ (-1, 1, 0, 0), (-1, -1, 2, 0), (-1, -1, -1, 3) \right\}$$

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Question #3.

find QR decomposition

$$A = \begin{bmatrix} 1 & 2 & 5 \\ -1 & 1 & -4 \\ 1 & 4 & -3 \\ 1 & -4 & 7 \\ 1 & 2 & 1 \end{bmatrix}$$

$$u_1 = (1, -1, -1, 1, 1), u_2 = (2, 1, 4, -4, -2) \\ u_3 = (5, -4, -3, 7, 1)$$

$$v_1 = u_1 = (-1, -1, -1, 1, 1) \Rightarrow q_1 = \left(\frac{1}{\sqrt{5}}, -\frac{1}{\sqrt{5}}, -\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right)$$

$$v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1 = (2, 1, 4, -4, -2) + \frac{5}{25} (1, -1, -1, 1, 1)$$

$$= (3, 0, 3, -3, 3)$$

$$v_3 = u_3 - \frac{\langle u_3, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle u_3, v_2 \rangle}{\|v_2\|^2} v_2 = (5, -4, -3, 7, 1) + \frac{(3, 0, 3, -3, 3)}{3}$$

$$-4(1, -1, -1, 1, 1) \\ v_3 = (2, 0, 2, 2, -2) \\ q_2 = \left(\frac{1}{2}, 0, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2} \right)$$

$$q_3 = \left(\frac{1}{2}, 0, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2} \right)$$

$$R = \begin{bmatrix} \sqrt{5} & -\sqrt{5} & 4\sqrt{5} \\ 0 & 6 & -2 \\ 0 & 0 & 4 \end{bmatrix}$$

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A: QR

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & 1 & -4 \\ -1 & 6 & -3 \\ 1 & -4 & 7 \\ 1 & 2 & 1 \end{bmatrix} \xrightarrow{R_2+R_1, R_3+R_1, R_4-R_1, R_5-R_1} \begin{bmatrix} 1/\sqrt{5} & 1/2 & 1/2 \\ -1/\sqrt{5} & 0 & 0 \\ -1/\sqrt{5} & 1/2 & 1/2 \\ 1/\sqrt{5} & -1/2 & 1/2 \\ 1/\sqrt{5} & 1/2 & -1/2 \end{bmatrix} \xrightarrow{R_2 \times (-1), R_3 \times (-1)} \begin{bmatrix} 1/\sqrt{5} & 1/2 & 1/2 \\ 0 & 6 & -2 \\ 0 & 0 & 4 \\ 1/\sqrt{5} & -1/2 & 1/2 \\ 1/\sqrt{5} & 1/2 & -1/2 \end{bmatrix}$$

Ans.

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Q) Question #4-

$$A = \begin{bmatrix} 9 & 4 \\ a & 1 \end{bmatrix}$$

$$\det(A) = 9 - 4a = 25$$

$$4a = -16$$

$$a = -4$$

Trace $a + 1 = 10$

Eigen vector for $\lambda = 5$

$$A - \lambda I = \begin{bmatrix} 9-5 & 4 \\ -4 & 1-5 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 4 \\ -4 & -4 \end{bmatrix}$$

Row Equation $4x + 4y = 0$

$$x = -y$$

so an eigenvector is $v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Since $A(1, -1)^T = 5(1, -1)^T$

2) matrix #2.

$$A = \begin{bmatrix} 10 & 5 \\ -5 & b \end{bmatrix}$$

$$\det(A) = 10b - 5 \times -5 = 10b + 25 = 25$$

$$b = 0$$

$$A = \begin{bmatrix} 10 & 5 \\ -5 & 0 \end{bmatrix}$$

EigenVector for $\lambda = 5$

$$A - \lambda I = \begin{bmatrix} 5 & 5 \\ -5 & -5 \end{bmatrix}$$

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again $\text{Int} S y = 0$
 $n = 0$

$$v \sim \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

& thus $A(1, -1)^T, S(1, -1)^T$

$$\begin{bmatrix} 9 & 4 \\ -4 & 1 \end{bmatrix} \text{ and } \begin{bmatrix} 10 & 5 \\ 5 & 0 \end{bmatrix}$$

each has $\text{tr}(A) = 10$, $\det(A) = 25$

characteristic polynomial $(\lambda - 5)^2$

Eigen vector space $\lambda = 5$

1d-span $\{(1, -1)^T\}$

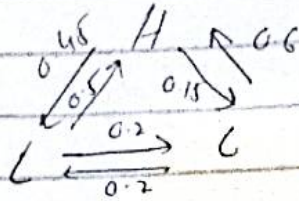
then matrices are non-diagonalizable

Names / Roll No.
Question # 1

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(a) Transition matrix: $H \begin{bmatrix} 0.4 & 0.45 & 0.15 \\ 0.5 & 0.3 & 0.2 \\ 0.6 & 0.2 & 0.2 \end{bmatrix}$

(b) Graph



(c) Given $\lambda[H][H] = 0.4$

(d) $\lambda[L][L] = 0.6$ given.

four days (each form)

$$0.6 \times 0.6 \times 0.6 \times 0.6 = (0.6)^4$$

$$= 0.1296$$

(e) find $A_{n \times n}$ given vector.

mean $A_{n \times n} \lambda n \Rightarrow \lambda 1$

$[\lambda I - A] = [I - A]$

$$\begin{bmatrix} 0.6 & -0.45 & -0.15 \\ -0.5 & 0.7 & -0.2 \\ -0.6 & 0.2 & 0.8 \end{bmatrix}$$

$$\frac{R_1}{0.6} \begin{bmatrix} 1 & -0.75 & -0.25 \\ -0.8 & 0.7 & -0.2 \\ -0.6 & 0.2 & 0.8 \end{bmatrix} \begin{matrix} R_2 + 0.5R_1 \\ R_3 + 0.6R_1 \end{matrix}$$

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$$\begin{bmatrix} 1 & -0.75 & -0.25 \\ 0 & 0.325 & -0.325 \\ 0 & -0.65 & 0.65 \end{bmatrix}$$

$$R_2 + 2R_1 \quad \begin{bmatrix} 1 & -0.75 & -0.25 \\ 0 & 0 & 0 \\ 0 & -0.65 & 0.65 \end{bmatrix} \quad R_2 \sim R_3 \quad \begin{bmatrix} 1 & -0.75 & -0.25 \\ 0 & -0.65 & 0.65 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{matrix} R_2 \\ -0.65 \end{matrix} \begin{bmatrix} 1 & -0.75 & -0.25 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \quad R_1 + 0.75R_2 \quad \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$n = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$$

